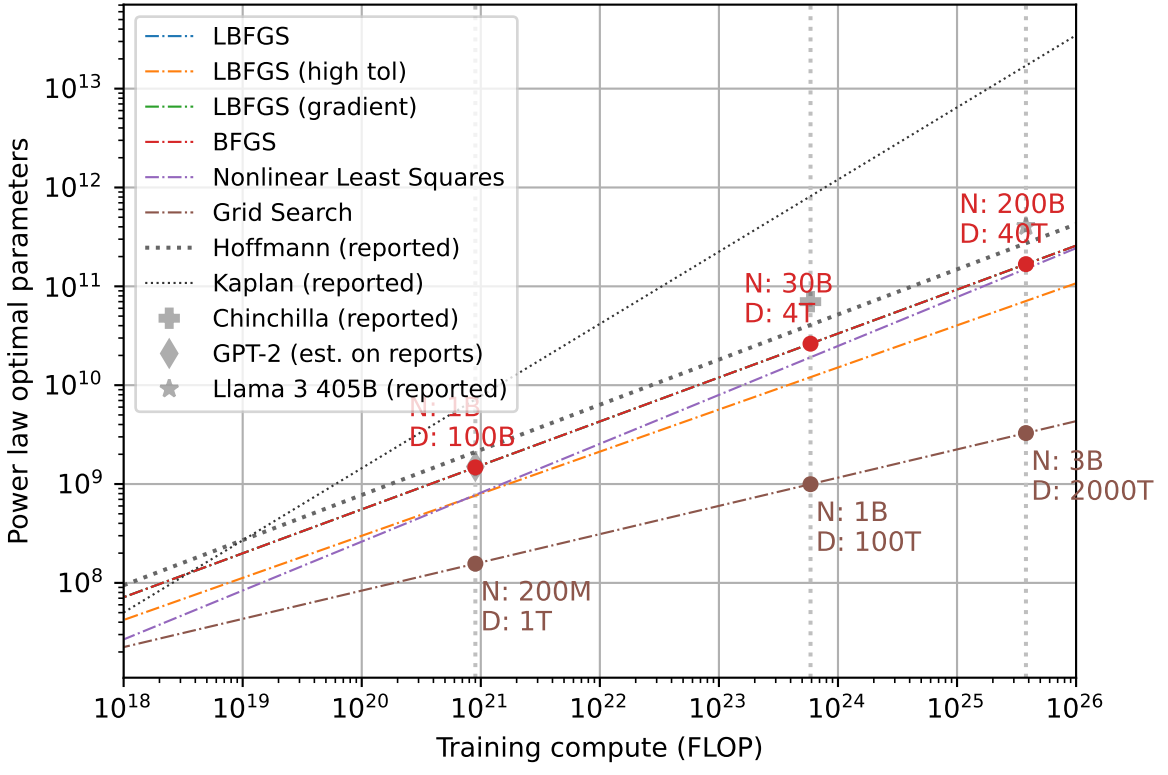
LBFGS
LBFGS (high tol)
LBFGS (gradient)
BFGS
Nonlinear Least Squares
Grid Search
Hoffmann (reported)
Kaplan (reported)
Chinchilla (reported)
GPT-2 (est. on reports)
Llama 3 405B (reported)
N: 1B
D: 100B
N: 30B
D: 4T
N: 200B
D: 40T
N: 200M
D: 1T
N: 1B
D: 100T
N: 3B
D: 2000T
Power law optimal parameters
Training compute (FLOP)
10^{8}
10^{9}
10^{10}
10^{11}
10^{12}
10^{13}
10^{18}
10^{19}
10^{20}
10^{21}
10^{22}
10^{23}
10^{24}
10^{25}
10^{26}